Supplementary Material

Granular ball local consensus for graph-free node classification under label noise

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This supplement provides extended experimental results, implementation details, and theoretical analyses supporting the main paper. It includes paired statistical tests, target-construction checks, additional datasets, full derivations, and parameter-sensitivity analyses.

Appendix A. Results of the main comparisons under all noise settings

All accuracies in this supplement are percentages, reported as the mean ± sample standard deviation over five paired seeds. Bold marks the highest mean within each dataset and noise setting; in Table C.1, the highest mean is marked separately for each validation protocol.

Appendix A.1. Cora

Table A.1: Test accuracy on Cora under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	54.96±2.22	44.24±2.62	32.51±3.38	42.83±2.19	45.33±2.22
Small-loss	59.84±1.29	51.95±1.97	36.72±1.09	46.99±2.77	51.82±0.94
Co-teaching	58.62±1.97	46.83±1.78	33.29±3.48	44.26±2.30	47.35±2.24
JoCoR	60.12±2.25	51.72±1.30	36.92±2.14	45.91±2.38	52.26±1.01
SCE	53.57±2.17	43.60±1.86	32.66±3.78	41.40±2.04	43.29±4.35
ELR	52.95±2.38	44.28±1.38	32.82±3.51	41.83±1.75	45.93±2.26
LogitClip	58.63±1.35	47.50±1.50	34.44±2.67	44.90±1.21	48.09±1.73
Random size-matched grouping	53.57±1.99	43.31±1.40	32.18±2.54	39.28±1.34	43.34±1.70
GB-LCSS	**63.25±2.46**	**57.49±0.76**	**41.86±2.30**	**51.45±2.80**	**56.84±1.67**

Appendix A.2. CiteSeer

Table A.2: Test accuracy on CiteSeer under all five noise settings.

Method	**Uniform 0.2**	**Uniform 0.4**	**Uniform 0.6**	**Pair 0.4**	**Instance 0.4**
MLP+CE	55.10±1.17	43.50±3.02	33.35±1.36	45.15±1.67	44.33±2.04
Small-loss	59.26±1.94	49.70±3.07	37.73±2.86	48.16±2.56	50.75±3.33
Co-teaching	57.05±0.82	46.24±2.89	34.19±2.92	46.25±2.29	46.81±1.68
JoCoR	59.36±1.43	49.91±3.37	39.07±2.47	48.63±2.35	50.86±2.45
SCE	52.12±3.28	42.09±3.47	30.59±2.83	42.22±1.56	40.92±4.23
ELR	53.62±1.48	43.17±3.95	30.80±2.70	42.69±2.43	41.94±3.27
LogitClip	58.75±1.44	46.57±3.52	34.47±1.69	45.78±1.06	47.41±2.64
Random size-matched grouping	55.13±1.85	42.08±2.88	30.36±1.88	43.92±1.67	43.07±2.00
GB-LCSS	**63.92±1.12**	**55.24±2.04**	**42.02±3.14**	**51.70±2.14**	**55.58±2.54**

Appendix A.3. CoraFull

Table A.3: Test accuracy on CoraFull under all five noise settings.

Method	**Uniform 0.2**	**Uniform 0.4**	**Uniform 0.6**	**Pair 0.4**	**Instance 0.4**
MLP+CE	41.11±0.46	31.98±0.98	21.16±0.81	28.14±1.17	31.33±0.85
Small-loss	41.58±0.94	33.04±0.50	21.51±0.98	30.15±1.11	32.90±0.87
Co-teaching	41.70±1.05	34.39±0.24	22.84±1.38	30.28±0.52	33.39±0.90
JoCoR	43.33±0.70	35.70±0.32	24.20±0.98	31.45±0.80	35.17±0.51
SCE	37.93±0.72	29.29±0.45	17.79±1.33	26.98±0.51	29.68±1.06
ELR	41.08±0.53	32.14±0.67	21.32±0.98	28.91±1.44	31.52±0.83
LogitClip	41.24±0.78	34.69±1.17	24.62±0.72	31.23±1.84	34.19±0.17
Random size-matched grouping	41.02±0.92	32.30±0.85	21.41±0.84	28.49±0.78	31.43±0.79
GB-LCSS	**43.46±0.57**	**37.39±0.51**	**27.43±0.89**	**34.46±1.23**	**36.95±0.53**

Appendix A.4. Coauthor-CS

Table A.4: Test accuracy on Coauthor-CS under all five noise settings.

Method	**Uniform 0.2**	**Uniform 0.4**	**Uniform 0.6**	**Pair 0.4**	**Instance 0.4**
MLP+CE	91.38±0.54	88.42±0.62	80.03±1.91	75.91±1.34	87.44±1.15
Small-loss	91.50±0.49	89.06±0.88	81.65±0.91	80.13±1.50	88.97±1.02
Co-teaching	91.70±0.29	88.80±0.97	81.88±0.72	79.06±1.30	88.95±0.97
JoCoR	**91.81±0.25**	89.38±0.61	82.58±0.41	81.02±1.62	89.17±0.85

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Table A.4 (continued)

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
SCE	90.63±0.47	88.37±0.53	79.43±1.45	79.77±2.51	87.99±0.56
ELR	91.21±0.60	88.29±0.73	80.54±1.28	78.03±1.48	87.78±0.94
LogitClip	91.32±0.43	88.40±0.21	79.96±1.64	81.57±0.53	88.60±0.75
Random size-matched grouping	90.26±0.70	87.46±1.07	78.78±1.82	75.99±0.60	86.87±1.56
GB-LCSS	91.54±0.26	**90.71±0.59**	**87.19±0.69**	**86.24±1.18**	**90.51±0.43**

Appendix A.5. Amazon-Photo

Table A.5: Test accuracy on Amazon-Photo under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	82.34±1.39	73.90±1.80	58.60±3.77	62.61±4.31	69.13±3.00
Small-loss	84.96±0.58	77.25±1.62	61.40±3.72	66.70±5.73	73.00±1.25
Co-teaching	85.21±0.51	78.21±1.00	63.29±5.00	67.24±6.97	73.59±1.19
JoCoR	85.60±1.26	79.57±0.53	66.48±4.43	66.94±6.36	74.80±1.79
SCE	84.74±0.94	76.08±3.11	57.82±4.07	66.51±6.96	73.08±2.54
ELR	82.31±1.98	73.93±2.12	59.32±4.53	66.07±5.25	69.41±2.70
LogitClip	83.78±1.02	74.98±2.38	62.42±4.37	68.52±4.20	72.90±1.05
Random size-matched grouping	81.03±1.23	71.45±3.29	58.30±4.17	62.56±4.24	67.56±2.89
GB-LCSS	**86.07±1.05**	**82.12±1.08**	**71.45±3.75**	**73.89±3.73**	**79.59±0.88**

Appendix B. Paired statistical tests

For each dataset and noise setting, we compute the accuracy difference between GB-LCSS and the reference method on each of the five paired seeds. Table B.1 reports the mean difference and an unadjusted 95% confidence interval in percentage points. "Wins (seeds)" counts the seeds on which GB-LCSS has higher accuracy. Two-sided paired t-test p-values are Holm-adjusted across the 25 comparisons with each reference method. The differences remain significant after correction in 24 of the 25 settings against MLP+CE and in 11 against JoCoR. GB-LCSS also exceeds JoCoR on all five seeds in 22 of the 25 settings.

Table B.1: Paired statistics for GB-LCSS against the three reference methods.

Dataset	Noise	Reference method	Mean difference (pp)	95% CI (pp)	Wins (seeds)	Adjusted p
Cora	Uniform 0.2	Random grouping	+9.68	[+7.16, +12.20]	5/5	0.006
Cora	Uniform 0.2	MLP+CE	+8.30	[+5.59, +11.00]	5/5	0.011
Cora	Uniform 0.2	JoCoR	+3.13	[+0.55, +5.72]	5/5	0.112
Cora	Uniform 0.4	Random grouping	+14.17	[+12.71, +15.63]	5/5	< 0.001
Cora	Uniform 0.4	MLP+CE	+13.25	[+10.81, +15.68]	5/5	0.002
Cora	Uniform 0.4	JoCoR	+5.77	[+3.60, +7.95]	5/5	0.031
Cora	Uniform 0.6	Random grouping	+9.68	[+6.39, +12.96]	5/5	0.010
Cora	Uniform 0.6	MLP+CE	+9.35	[+4.54, +14.16]	5/5	0.011
Cora	Uniform 0.6	JoCoR	+4.94	[+1.79, +8.09]	5/5	0.082
Cora	Pair 0.4	Random grouping	+12.17	[+7.23, +17.11]	5/5	0.012
Cora	Pair 0.4	MLP+CE	+8.62	[+5.74, +11.50]	5/5	0.011
Cora	Pair 0.4	JoCoR	+5.54	[+2.90, +8.18]	5/5	0.060
Cora	Instance 0.4	Random grouping	+13.51	[+12.08, +14.94]	5/5	< 0.001
Cora	Instance 0.4	MLP+CE	+11.51	[+9.50, +13.53]	5/5	0.002
Cora	Instance 0.4	JoCoR	+4.58	[+2.27, +6.89]	5/5	0.062
CiteSeer	Uniform 0.2	Random grouping	+8.79	[+6.87, +10.71]	5/5	0.004
CiteSeer	Uniform 0.2	MLP+CE	+8.81	[+7.00, +10.63]	5/5	0.003
CiteSeer	Uniform 0.2	JoCoR	+4.56	[+2.87, +6.26]	5/5	0.031
CiteSeer	Uniform 0.4	Random grouping	+13.16	[+11.26, +15.06]	5/5	< 0.001
CiteSeer	Uniform 0.4	MLP+CE	+11.74	[+9.99, +13.50]	5/5	0.001
CiteSeer	Uniform 0.4	JoCoR	+5.34	[+2.51, +8.16]	5/5	0.062
CiteSeer	Uniform 0.6	Random grouping	+11.66	[+9.60, +13.72]	5/5	0.002
CiteSeer	Uniform 0.6	MLP+CE	+8.67	[+4.92, +12.42]	5/5	0.011
CiteSeer	Uniform 0.6	JoCoR	+2.95	[+1.09, +4.81]	5/5	0.082
CiteSeer	Pair 0.4	Random grouping	+7.78	[+5.04, +10.53]	5/5	0.010
CiteSeer	Pair 0.4	MLP+CE	+6.55	[+3.98, +9.13]	5/5	0.011
CiteSeer	Pair 0.4	JoCoR	+3.07	[+2.06, +4.07]	5/5	0.020
CiteSeer	Instance 0.4	Random grouping	+12.51	[+9.97, +15.05]	5/5	0.003
CiteSeer	Instance 0.4	MLP+CE	+11.25	[+8.48, +14.01]	5/5	0.006
CiteSeer	Instance 0.4	JoCoR	+4.72	[+3.63, +5.80]	5/5	0.007
CoraFull	Uniform 0.2	Random grouping	+2.44	[+1.48, +3.40]	5/5	0.012
CoraFull	Uniform 0.2	MLP+CE	+2.35	[+1.46, +3.25]	5/5	0.011
CoraFull	Uniform 0.2	JoCoR	+0.13	[-0.86, +1.12]	3/5	0.852

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Table B.1 (continued)

Dataset	Noise	Reference method	Mean difference (pp)	95% CI (pp)	Wins (seeds)	Adjusted p
CoraFull	Uniform 0.4	Random grouping	+5.09	[+4.32, +5.87]	5/5	0.001
CoraFull	Uniform 0.4	MLP+CE	+5.41	[+4.51, +6.32]	5/5	0.002
CoraFull	Uniform 0.4	JoCoR	+1.69	[+0.86, +2.52]	5/5	0.062
CoraFull	Uniform 0.6	Random grouping	+6.02	[+5.16, +6.89]	5/5	< 0.001
CoraFull	Uniform 0.6	MLP+CE	+6.27	[+5.52, +7.03]	5/5	< 0.001
CoraFull	Uniform 0.6	JoCoR	+3.23	[+1.89, +4.57]	5/5	0.042
CoraFull	Pair 0.4	Random grouping	+5.98	[+4.75, +7.20]	5/5	0.003
CoraFull	Pair 0.4	MLP+CE	+6.32	[+4.66, +7.98]	5/5	0.007
CoraFull	Pair 0.4	JoCoR	+3.02	[+2.10, +3.93]	5/5	0.017
CoraFull	Instance 0.4	Random grouping	+5.52	[+4.23, +6.80]	5/5	0.004
CoraFull	Instance 0.4	MLP+CE	+5.62	[+4.37, +6.86]	5/5	0.004
CoraFull	Instance 0.4	JoCoR	+1.78	[+1.22, +2.34]	5/5	0.018
Coauthor-CS	Uniform 0.2	Random grouping	+1.28	[+0.70, +1.85]	5/5	0.012
Coauthor-CS	Uniform 0.2	MLP+CE	+0.16	[-0.33, +0.65]	3/5	0.418
Coauthor-CS	Uniform 0.2	JoCoR	-0.27	[-0.41, -0.13]	0/5	0.062
Coauthor-CS	Uniform 0.4	Random grouping	+3.25	[+1.96, +4.55]	5/5	0.012
Coauthor-CS	Uniform 0.4	MLP+CE	+2.29	[+1.66, +2.93]	5/5	0.008
Coauthor-CS	Uniform 0.4	JoCoR	+1.33	[+1.07, +1.59]	5/5	0.004
Coauthor-CS	Uniform 0.6	Random grouping	+8.41	[+5.72, +11.10]	5/5	0.009
Coauthor-CS	Uniform 0.6	MLP+CE	+7.16	[+4.58, +9.75]	5/5	0.011
Coauthor-CS	Uniform 0.6	JoCoR	+4.61	[+3.37, +5.85]	5/5	0.011
Coauthor-CS	Pair 0.4	Random grouping	+10.25	[+8.67, +11.83]	5/5	0.001
Coauthor-CS	Pair 0.4	MLP+CE	+10.32	[+8.97, +11.67]	5/5	< 0.001
Coauthor-CS	Pair 0.4	JoCoR	+5.22	[+3.82, +6.62]	5/5	0.011
Coauthor-CS	Instance 0.4	Random grouping	+3.65	[+2.12, +5.17]	5/5	0.012
Coauthor-CS	Instance 0.4	MLP+CE	+3.07	[+2.15, +4.00]	5/5	0.009
Coauthor-CS	Instance 0.4	JoCoR	+1.34	[+0.76, +1.92]	5/5	0.045
Amazon-Photo	Uniform 0.2	Random grouping	+5.03	[+3.82, +6.25]	5/5	0.005
Amazon-Photo	Uniform 0.2	MLP+CE	+3.72	[+2.63, +4.81]	5/5	0.009
Amazon-Photo	Uniform 0.2	JoCoR	+0.46	[-0.99, +1.92]	3/5	0.852

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Table B.1 (continued)

Dataset	Noise	Reference method	Mean difference (pp)	95% CI (pp)	Wins (seeds)	Adjusted p
Amazon-Photo	Uniform 0.4	Random grouping	+10.66	[+6.55, +14.78]	5/5	0.012
Amazon-Photo	Uniform 0.4	MLP+CE	+8.22	[+5.17, +11.26]	5/5	0.011
Amazon-Photo	Uniform 0.4	JoCoR	+2.55	[+1.02, +4.07]	5/5	0.078
Amazon-Photo	Uniform 0.6	Random grouping	+13.15	[+9.63, +16.67]	5/5	0.006
Amazon-Photo	Uniform 0.6	MLP+CE	+12.85	[+8.39, +17.31]	5/5	0.011
Amazon-Photo	Uniform 0.6	JoCoR	+4.96	[+0.60, +9.32]	5/5	0.112
Amazon-Photo	Pair 0.4	Random grouping	+11.32	[+8.02, +14.63]	5/5	0.008
Amazon-Photo	Pair 0.4	MLP+CE	+11.28	[+9.10, +13.46]	5/5	0.003
Amazon-Photo	Pair 0.4	JoCoR	+6.95	[+1.90, +12.00]	5/5	0.094
Amazon-Photo	Instance 0.4	Random grouping	+12.03	[+8.45, +15.61]	5/5	0.008
Amazon-Photo	Instance 0.4	MLP+CE	+10.46	[+6.95, +13.98]	5/5	0.011
Amazon-Photo	Instance 0.4	JoCoR	+4.79	[+2.43, +7.15]	5/5	0.062

Random grouping preserves the number and sizes of the granular balls.

Appendix C. Sensitivity to noisy validation labels

We evaluate checkpoint selection with noisy validation labels under uniform 0.4 training-label noise. The data splits and noisy training labels are unchanged, and validation labels are independently corrupted at the same rate using a separate random seed. Test labels remain clean. GB-LCSS and random size-matched grouping use the targets from Section 3.3 of the main paper; GSDN and PosteL use the implementations described in Section 4.1.3 and Appendix F.

Table C.1: Test accuracy under clean and noisy validation labels.

Dataset	Method	Clean validation	Noisy validation
Cora	GB-LCSS	**57.49±0.76**	**56.81±1.55**
Cora	Random size-matched grouping	43.31±1.40	43.12±2.07
Cora	GSDN	55.97±4.29	54.81±4.70
Cora	MLP+CE	44.24±2.62	43.30±3.65
Cora	JoCoR	51.72±1.30	50.77±1.98

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Table C.1 (continued)

Dataset	Method	Clean validation	Noisy validation
Cora	PosteL	47.09±1.81	46.81±1.94
CiteSeer	GB-LCSS	**55.24±2.04**	**54.99±2.18**
CiteSeer	Random size-matched grouping	42.08±2.88	42.65±2.49
CiteSeer	GSDN	50.36±3.39	50.60±1.84
CiteSeer	MLP+CE	43.50±3.02	43.06±2.96
CiteSeer	JoCoR	49.91±3.37	50.67±1.07
CiteSeer	PosteL	45.84±2.48	46.27±1.76
CoraFull	GB-LCSS	**37.39±0.51**	**36.90±0.81**
CoraFull	Random size-matched grouping	32.30±0.85	32.03±0.77
CoraFull	GSDN	35.80±0.66	36.13±0.69
CoraFull	MLP+CE	31.98±0.98	31.99±0.85
CoraFull	JoCoR	35.70±0.32	35.57±1.04
CoraFull	PosteL	36.51±1.01	35.74±1.14

Using noisy validation labels changes the test accuracy of GB-LCSS by at most 0.68 percentage points across the three datasets. GB-LCSS remains above random size-matched grouping by 13.69, 12.35, and 4.87 points on Cora, CiteSeer, and CoraFull, respectively, and above MLP+CE by 13.51, 11.93, and 4.91 points. It also retains the highest mean accuracy among the six methods on all three datasets (Table C.1).

Appendix D. Analysis of target construction

Table D.1 summarizes the leave-one-out and clean-label checks. Excluding each node's own label from its local vote improves mean accuracy on all five datasets when averaged over the five noise settings. In the least favorable individual setting, excluding the vote lowers accuracy by 0.49 points.

With clean training labels ($\eta = 0$) and all other settings unchanged, GB-LCSS has higher mean accuracy than MLP+CE on CiteSeer and lower mean accuracy on the other four datasets. Targets can change only in balls whose training nodes belong to more than one class.

When two or more classes receive the same highest vote count, GB-LCSS retains the empirical vote distribution as the consensus, $\mathbf{q}_n = \mathbf{r}_n$ (Eq. (10)). Across the five datasets, five noise settings, and five seeds, these ties affect 12,788 of the 259,000 training-node targets (4.94%).

Appendix E. Results of graph-based soft-target methods under all noise settings

Table E.1 reports GB-LCSS, PosteL, ALS, and SALS on the five main datasets under all five noise settings, using the common splits, noise realizations, and graph-free inference protocol (Appendix F). GB-LCSS has the highest mean in 24 of the 25

Table D.1: Accuracy differences in the leave-one-out and clean-label checks.

Dataset	**Excluding minus including self-vote**	**GB-LCSS minus MLP+CE**
	Five noise settings	Clean labels
Cora	+0.96	-0.24
CiteSeer	+1.03	+2.00
CoraFull	+0.09	-0.38
Coauthor-CS	+0.08	-0.87
Amazon-Photo	+0.29	-1.75

Differences are in percentage points. The leave-one-out comparison averages paired differences over five seeds and five noise settings. The clean-label comparison averages paired differences over five seeds at zero training-label noise.

settings; the exception is CoraFull under uniform 0.2, where PosteL and SALS are higher.

Table E.1: Test accuracy of graph-based soft-target methods under all five noise settings.

Dataset	**Noise setting**	**GB-LCSS**	**PosteL**	**ALS**	**SALS**
Cora	Uniform 0.2	**63.25±2.46**	58.36±1.64	55.98±1.42	58.23±1.67
Cora	Uniform 0.4	**57.49±0.76**	47.09±1.81	45.20±1.24	48.27±2.50
Cora	Uniform 0.6	**41.86±2.30**	32.33±2.37	32.72±2.54	35.34±1.74
Cora	Pair 0.4	**51.45±2.80**	44.36±0.65	41.63±3.61	44.16±1.52
Cora	Instance 0.4	**56.84±1.67**	46.62±2.17	45.73±1.34	48.54±1.17
CiteSeer	Uniform 0.2	**63.92±1.12**	58.11±1.11	56.58±2.04	58.36±1.07
CiteSeer	Uniform 0.4	**55.24±2.04**	45.84±2.48	43.77±2.76	47.20±1.69
CiteSeer	Uniform 0.6	**42.02±3.14**	31.88±0.86	32.00±1.57	35.16±2.58
CiteSeer	Pair 0.4	**51.70±2.14**	45.06±1.60	45.72±2.41	45.63±2.70
CiteSeer	Instance 0.4	**55.58±2.54**	46.02±2.54	43.75±2.18	47.51±2.50
CoraFull	Uniform 0.2	43.46±0.57	**44.54±0.31**	41.72±0.22	44.43±0.75
CoraFull	Uniform 0.4	**37.39±0.51**	36.51±1.01	31.91±1.05	37.06±0.78
CoraFull	Uniform 0.6	**27.43±0.89**	23.90±1.16	20.64±0.61	26.10±1.07
CoraFull	Pair 0.4	**34.46±1.23**	32.62±1.71	29.83±0.92	32.85±1.68
CoraFull	Instance 0.4	**36.95±0.53**	35.99±0.51	31.15±1.18	36.58±0.74
Coauthor-CS	Uniform 0.2	**91.54±0.26**	89.01±0.40	91.13±0.57	91.08±0.49
Coauthor-CS	Uniform 0.4	**90.71±0.59**	84.74±1.51	88.55±0.86	88.85±0.82
Coauthor-CS	Uniform 0.6	**87.19±0.69**	74.61±1.95	79.36±1.67	82.99±0.95
Coauthor-CS	Pair 0.4	**86.24±1.18**	78.45±2.28	74.99±2.24	77.85±1.51
Coauthor-CS	Instance 0.4	**90.51±0.43**	85.72±1.35	87.53±1.27	88.33±0.66

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Table E.1 (continued)

Dataset	Noise setting	GB-LCSS	PosteL	ALS	SALS
Amazon-Photo	Uniform 0.2	**86.07±1.05**	84.37±0.96	82.63±1.27	84.81±1.20
Amazon-Photo	Uniform 0.4	**82.12±1.08**	80.59±0.94	73.41±2.75	77.86±0.79
Amazon-Photo	Uniform 0.6	**71.45±3.75**	64.49±2.99	58.14±1.78	66.29±3.62
Amazon-Photo	Pair 0.4	**73.89±3.73**	67.56±3.57	62.82±6.11	64.83±3.54
Amazon-Photo	Instance 0.4	**79.59±0.88**	77.23±2.77	68.81±2.81	72.73±1.78

Appendix F. Implementation details and key settings

For seeds 0–4, the NoisyGL percentage-split routine randomly assigns nodes to disjoint training, validation, and test sets of sizes $\lfloor 0.2N \rfloor$, $\lfloor 0.1N \rfloor$, and $\lfloor 0.7N \rfloor$. Sampling is independent of class labels. Representation learning and partitioning use all nodes as unlabeled transductive context.

Table F.1 identifies the implementation used for each method and the components retained under the common protocol. All methods share the data splits and noise realizations and deploy an attribute-only MLP. Table F.2 lists their method-specific settings and sources.

Table F.1: Implementations of graph-based and distillation methods.

Method	Implementation	Components and adaptations
PosteL	Authors' target-generation code	Posterior soft-target construction; iterative pseudo-labeling disabled
ALS	Mechanism adaptation	Adaptive label smoothing, pacing, and KL term
SALS	Mechanism adaptation	Structure-aware label smoothing
Graph-MLP	Adapted to the common MLP	Multi-hop NContrast objective
GSDN	Mechanism adaptation	Neighborhood self-distillation at the feature and label levels
GCN(X)+CE teacher	Implemented for this study	GCN-to-MLP response distillation
FF-G2M	Reimplemented with the common GCN and MLP	Low- and high-frequency distillation
KRD	Mechanism adaptation	Reliability-based knowledge selection
VQGraph	Mechanism adaptation	Soft-codebook distillation

The optimizer and MLP settings shared across methods are given in Section 4.1.4 of the main paper.

Table F.2: Key method-specific settings used in the main comparisons.

Method	Key settings	Parameter selection
Small-loss	Discarded fraction (forget rate) increases linearly to the synthetic noise rate over 10 epochs; exponent 1.0	Fixed schedule across datasets; final rate given by synthetic noise level
Co-teaching	Two independent MLPs; same forget-rate schedule as Small-loss	NoisyGL configuration; Small-loss schedule
JoCoR	Two independent MLPs; same forget-rate schedule as Small-loss; consistency weight 0.1	NoisyGL configuration; Small-loss schedule
SCE	Cross-entropy weight 0.1; reverse-cross-entropy weight 1.0	NoisyGL configuration
ELR	Moving-average coefficient 0.7; regularization weight 1.0	Fixed across datasets
LogitClip	Logit-clipping threshold 1.0	Fixed across datasets
PosteL	Soft-label ratio 0.8; smoothing ratio 0.4; degree cutoff 1; temperature 1.0; iterative pseudo-labeling disabled	Authors' target code under the common training protocol
ALS	2 propagation steps; residual coefficient 0.5; KL weight 1e-3; exponential pacing; maximum smoothing weight 0.1	Fixed across datasets
SALS	epsilon = 0.4; gamma = 0.8; unlabeled neighborhoods use the uniform distribution	Fixed across datasets
Graph-MLP	Graph batch size 512; neighborhood-contrastive weight 0.5; temperature 0.5	Fixed across datasets
GSDN	Graph batch size 512; weights of 0.8 at both feature and label levels; negative-constraint weight 0.1; initial mixup coefficient 0.5	Fixed across datasets
GB-LCSS	$B_{\max} = 64$, $n_{\min} = 32$, $\delta = 0.05$, $s_{\min} = 2$, $\beta = 0.5$; current node excluded from voting; ties keep the empirical distribution; 2-means uses Euclidean distance, k-means++, and 10 restarts	Established through iterative method development; used unchanged in the main comparisons
GCN(X)+CE teacher	Two-layer GCN teacher and two-layer MLP student, both width 64; 200 epochs; temperature 1.0	Shared settings for teacher and student training
FF-G2M	Two-layer GCN teacher, width 64; low- and high-frequency distillation weights 0.4; temperature 1.5	Fixed across datasets
KRD	4 feature perturbations, standard deviation 0.01; top 50% of nodes retained by reliability; temperature 1.0	Fixed across datasets

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Table F.2 (continued)

Method	Key settings	Parameter selection
VQGraph	128 soft codes; codebook temperature 1.0; token, node, and edge weights of 1.0, 0.001, and 0.03; 4096 edges sampled per epoch	Fixed across datasets
GCN teacher trained on GB-LCSS targets	Two-layer GCN(X) teacher, width 64; GB-LCSS targets from Eq. (12); student temperature 1.0; student fits teacher predictions over all nodes	Shared settings for teacher and student training; GB-LCSS targets

Table F.3 lists the GRACE settings for each dataset. Each dataset uses one frozen representation across noise settings and downstream seeds.

Table F.3: GRACE settings for the five main datasets.

Dataset	Dimensions	Learning rate	Epochs	Edge dropping	Feature masking	Temperature
Cora	(128, 128)	5×10^{-4}	200	(0.2, 0.4)	(0.3, 0.4)	0.4
CiteSeer	(256, 256)	10^{-3}	200	(0.2, 0.0)	(0.3, 0.2)	0.9
CoraFull	(256, 256)	10^{-3}	1000	(0.1, 0.4)	(0.1, 0.0)	0.7
Coauthor-CS	(256, 256)	10^{-3}	1000	(0.1, 0.4)	(0.1, 0.0)	0.7
Amazon-Photo	(256, 256)	10^{-3}	1500	(0.4, 0.1)	(0.0, 0.2)	0.7

Public GRACE configurations provide presets for Cora, CiteSeer, DBLP, and PubMed. We use the corresponding presets for Cora and CiteSeer. CoraFull and Coauthor-CS use the DBLP preset, and Amazon-Photo uses the PubMed preset. In the Dimensions column, each pair gives the output dimensions of the encoder and projection head, respectively. The two rates in each augmentation column apply to the first and second augmented graph views. All configurations use a two-layer GCN with weight decay 10^{-5}.

Appendix G. Results on additional datasets

The six additional datasets were first evaluated under uniform 0.4 noise. CiteSeerFull was subsequently evaluated under the remaining four noise settings as an exploratory follow-up. Table G.1 summarizes the statistics of the six datasets; data processing and edge homophily follow Table 2 of the main paper.

Table G.1: Statistics of the six additional datasets.

Dataset	Nodes	Undirected edges	Features	Classes	Training nodes	Edge homophily
PubMed	19,717	44,324	500	3	3,943	0.802
Amazon-Computers	13,752	245,861	767	10	2,750	0.777
Cora_ML	2,995	8,158	2,879	7	599	0.789
Attributed-Wiki	2,405	11,596	4,973	17	481	0.610

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Table G.1 (continued)

Dataset	Nodes	Undirected edges	Features	Classes	Training nodes	Edge homophily
CiteSeerFull	4,230	5,337	602	6	846	0.949
Coauthor-Physics	34,493	247,962	8,415	5	6,898	0.931

Table G.2 compares GB-LCSS with the strongest of the seven noisy-label baselines in Table 3 of the main paper and with random size-matched grouping under uniform 0.4 noise. For each dataset, the best baseline is identified by its five-seed mean accuracy. Differences are GB-LCSS minus the corresponding method, in percentage points; "Wins (seeds)" counts the paired seeds on which GB-LCSS exceeds the listed best baseline. The same definitions apply to Table G.3.

Table G.2: Uniform 0.4 results on the six additional datasets.

Dataset	GB-LCSS	Best baseline	Baseline accuracy	Difference vs. baseline	Wins (seeds)	Random grouping	Difference vs. random
PubMed	72.60±0.66	JoCoR	**77.84±1.10**	-5.24	0/5	63.04±4.49	+9.56
Amazon-Computers	76.20±0.56	JoCoR	**76.81±0.85**	-0.60	1/5	70.49±1.08	+5.71
Cora_ML	**56.26±2.61**	Small-loss	55.56±3.42	+0.70	3/5	47.81±3.34	+8.45
Attributed-Wiki	65.17±1.42	LogitClip	**67.83±1.51**	-2.66	0/5	59.83±2.13	+5.34
CiteSeerFull	**60.94±1.08**	JoCoR	59.48±1.60	+1.46	4/5	46.69±2.74	+14.25
Coauthor-Physics	**93.13±0.25**	JoCoR	93.09±0.28	+0.04	3/5	88.80±2.12	+4.33

GB-LCSS has a higher mean accuracy than random size-matched grouping on all six datasets, by 4.33–14.25 percentage points, and exceeds the strongest noisy-label baseline on three. The difference from the strongest baseline ranges from −5.24 points on PubMed to +1.46 points on CiteSeerFull. Table G.3 reports the results for CiteSeerFull under all five noise settings: GB-LCSS has a mean accuracy at least as high as that of the strongest baseline in four settings, whereas the strongest baseline is higher by 4.20 points under uniform 0.6.

Table G.3: CiteSeerFull results under all five noise settings.

Noise setting	GB-LCSS	Best baseline	Baseline accuracy	Difference vs. baseline	Wins (seeds)	Random grouping
Uniform 0.2	**70.71±0.65**	Small-loss	67.47±1.98	+3.24	5/5	58.70±1.88
Uniform 0.4	**60.94±1.08**	JoCoR	59.48±1.60	+1.46	4/5	46.69±2.74
Uniform 0.6	42.45±3.08	JoCoR	**46.65±3.70**	-4.20	0/5	33.13±3.53
Pair 0.4	**55.02±4.07**	JoCoR	51.12±5.88	+3.89	4/5	43.86±3.51
Instance 0.4	**61.49±3.01**	JoCoR	59.92±4.19	+1.57	3/5	47.51±2.21

Appendix H. Analysis of local-consensus supervision

Appendix H.1 proves the local-vote error bound in Proposition 1. Appendix H.2 derives the target-error identity in Eq. (16) and the sufficient condition in Corollary 1 for reducing expected target error. As in Section 3.4, the graph, representation, ball assignments, training-node set, and clean labels are fixed, and randomness comes only from independent symmetric label corruption at rate η.

Appendix H.1. Reliability of the local vote

Consider a training node v_n with $s_n = |U_n| \geq s_{\min}$ other training nodes voting in its ball. The clean class proportions $a_{n,i}$ and the margin Δ_n are defined in Eq. (14). As in Proposition 1, assume $\Delta_n > 0$ and $0 \leq \eta < (C-1)/C$. Each voting label retains its clean class with probability $1-\eta$ and changes to any particular incorrect class with probability $\eta/(C-1)$. With $\gamma = 1 - C\eta/(C-1) > 0$, the expected proportion of observed votes for class i is

$$\mathbb{E}[r_{n,i}] = (1-\eta)a_{n,i} + \frac{\eta}{C-1}(1-a_{n,i}) = \gamma a_{n,i} + \frac{\eta}{C-1}. \tag{H.1}$$

The expected vote difference between the true class of node v_n and any competing class $i \neq y_n$ is therefore

$$\mathbb{E}[r_{n,y_n} - r_{n,i}] = \gamma(a_{n,y_n} - a_{n,i}) \geq \gamma\Delta_n. \tag{H.2}$$

Symmetric noise therefore moves the expected vote proportions toward $1/C$ and reduces the margin over the closest competitor from Δ_n to $\gamma\Delta_n$. For $\gamma > 0$, the true class retains its lead in expectation. At $\eta = (C-1)/C$, all observed classes are equally likely.

For a competing class $i \neq y_n$, define $\xi_\ell = \mathbb{I}(z_\ell = y_n) - \mathbb{I}(z_\ell = i)$ for each voting node $v_\ell \in U_n$. This variable equals $+1$ for a vote for the true class of v_n, -1 for a vote for class i, and 0 otherwise. The variables are independent across nodes and lie in $[-1, 1]$, and their average is $r_{n,y_n} - r_{n,i}$. Using the expected difference in Eq. (H.2), Hoeffding's inequality [H1] bounds the probability that class i ties or exceeds the true class:

$$\Pr(r_{n,i} \geq r_{n,y_n}) \leq \exp\left[-\frac{s_n\gamma^2(a_{n,y_n} - a_{n,i})^2}{2}\right] \leq \exp\left(-\frac{s_n\gamma^2\Delta_n^2}{2}\right). \tag{H.3}$$

Applying the union bound to the $C-1$ competing classes and capping the result at one gives Eq. (15), proving Proposition 1.

Equation (10) preserves the most-voted classes, so the same bound applies to $\mathbf{q}_n$. Because Proposition 1 requires $\Delta_n > 0$, Appendix I.2 reports the proportion of training nodes satisfying this condition.

Appendix H.2. Error of the consensus-weighted target

For a training node v_n with $|U_n| \geq s_{\min}$, let $\mathbf{e}_{z_n}$ and $\mathbf{e}_{y_n}$ be its observed and clean one-hot labels. The Brier score of a target $\mathbf{b}$ is $\mathrm{BS}_n(\mathbf{b}) = \|\mathbf{b} - \mathbf{e}_{y_n}\|_2^2$. For any fixed realization of the observed labels, substituting $\mathbf{t}_n = (1-\alpha_n)\mathbf{e}_{z_n} + \alpha_n\mathbf{q}_n$ and expanding the squared Euclidean distance gives

$$\|\mathbf{t}_n - \mathbf{e}_{y_n}\|_2^2 = (1-\alpha_n)^2\|\mathbf{e}_{z_n} - \mathbf{e}_{y_n}\|_2^2 + \alpha_n^2\|\mathbf{q}_n - \mathbf{e}_{y_n}\|_2^2 + 2\alpha_n(1-\alpha_n)\langle \mathbf{e}_{z_n} - \mathbf{e}_{y_n}, \mathbf{q}_n - \mathbf{e}_{y_n}\rangle. \tag{H.4}$$

Substituting $2\langle \mathbf{e}_{z_n} - \mathbf{e}_{y_n}, \mathbf{q}_n - \mathbf{e}_{y_n} \rangle = \|\mathbf{e}_{z_n} - \mathbf{e}_{y_n}\|_2^2 + \|\mathbf{q}_n - \mathbf{e}_{y_n}\|_2^2 - \|\mathbf{e}_{z_n} - \mathbf{q}_n\|_2^2$ into Eq. (H.4) and collecting terms gives Eq. (16), relating the target error to the errors of the observed label and the local consensus. Nodes with fewer than $s_{\min}$ votes retain their observed labels and hence their original errors.

Two cases follow directly. If the observed label is correct, then $\mathbf{e}_{z_n} = \mathbf{e}_{y_n}$, $\mathrm{BS}_n(\mathbf{e}_{z_n}) = 0$, and Eq. (16) reduces to $\mathrm{BS}_n(\mathbf{t}_n) = \alpha_n^2 \, \mathrm{BS}_n(\mathbf{q}_n)$. If the observed label is corrupted, then $\mathbf{e}_{z_n}$ and $\mathbf{e}_{y_n}$ are distinct one-hot vectors, so $\mathrm{BS}_n(\mathbf{e}_{z_n}) = 2$, and $\mathrm{BS}_n(\mathbf{t}_n) < 2$ whenever $\alpha_n > 0$ and $\mathrm{BS}_n(\mathbf{q}_n) < 2$.

For the full training set, let $\mathcal{V}_{\mathrm{err}} = \{v_n \in \mathcal{V}_L : z_n \neq y_n\}$ denote the set of nodes with corrupted labels and $\mathcal{V}_{\mathrm{clean}} = \mathcal{V}_L \setminus \mathcal{V}_{\mathrm{err}}$ the set of nodes with correct labels. For a target matrix, BS denotes the mean of the node-wise scores BS_n over the labeled training nodes. With $\mathbf{Z}$ and $\mathbf{T}$ denoting the observed one-hot and soft-target matrices, respectively, the difference between their mean Brier scores is

$$\mathrm{BS}(\mathbf{Z}) - \mathrm{BS}(\mathbf{T}) = \frac{1}{|\mathcal{V}_L|} \sum_{v_n \in \mathcal{V}_{\mathrm{err}}} (2 - \mathrm{BS}_n(\mathbf{t}_n)) - \frac{1}{|\mathcal{V}_L|} \sum_{v_n \in \mathcal{V}_{\mathrm{clean}}} \mathrm{BS}_n(\mathbf{t}_n). \tag{H.5}$$

The first term measures the error removed on corrupted labels, and the second measures the error introduced on correct labels. Both are averaged over the full training set, and their difference is the net reduction reported in Table I.2. A target can have a lower Brier score while still assigning its largest probability to an incorrect class, because this score evaluates the full probability distribution.

For Corollary 1, fix the observed labels of all training nodes except v_n. The leave-one-out rule then keeps $\mathbf{q}_n$ and α_n fixed as the observed label of v_n varies. The observed one-hot label has squared error 0 when correct and 2 when corrupted, so $\mathbb{E}[\mathrm{BS}_n(\mathbf{e}_{z_n})] = 2\eta$, where the expectation is over the corruption of v_n's label. Taking this expectation in Eq. (16) and dropping its nonpositive final term gives

$$\mathbb{E}[\mathrm{BS}_n(\mathbf{t}_n)] \leq 2\eta(1 - \alpha_n) + \alpha_n \, \mathrm{BS}_n(\mathbf{q}_n). \tag{H.6}$$

This proves Eq. (17). When $\mathrm{BS}_n(\mathbf{q}_n) < 2\eta$ and $\alpha_n > 0$, the upper bound is strictly below 2η, the expected error of the observed one-hot label. The consensus-weighted target therefore has lower expected error, proving Corollary 1. Section 4.3.1 and Appendix I.2 evaluate these target errors empirically.

Reference for Appendix H. [H1] W. Hoeffding, Probability inequalities for sums of bounded random variables, J. Am. Stat. Assoc. 58 (1963) 13–30.

Appendix I. Parameter sensitivity and target-error decomposition

Appendix I.1. Parameter sensitivity and representation controls

We evaluate the grouping representations and target-construction parameters on the five main datasets under uniform 0.4 noise and seeds 0–4. Each variant uses the same data splits, noisy labels, initialization seeds, MLP, optimizer, 200 training epochs, and checkpoint-selection rule as the default configuration (Section 4.1). GRACE-based variants reuse the frozen representations from the main comparisons.

Each parameter variant in Table I.1 changes one parameter from its default value in Section 4.1.4. The ball assignments remain fixed when varying β or $s_{\min}$; $\beta = 0$ uses the empirical vote distribution as the consensus. Random size-matched grouping and the raw $\mathbf{X}$ and $\mathbf{A}^2\mathbf{X}$ representations are the controls described in Sections 4.3.1 and 4.3.3. Bold marks the highest mean in each column.

Table I.1: Parameter sensitivity and controls under uniform 0.4 noise.

Configuration	Cora	CiteSeer	CoraFull	Coauthor-CS	Amazon-Photo
Default	57.49±0.76	55.24±2.04	37.39±0.51	90.71±0.59	82.12±1.08
$\beta = 0$	54.85±2.01	52.81±3.62	37.11±0.32	89.97±0.58	79.26±1.90
$\beta = 0.25$	56.16±1.49	54.31±2.58	37.61±0.32	90.24±0.52	81.09±0.58
$\beta = 0.75$	57.63±1.14	56.23±1.79	36.98±0.31	90.76±0.37	**82.82±0.94**
$\beta = 1$	**57.65±1.05**	**56.98±2.07**	37.08±1.18	**91.01±0.31**	82.79±0.76
$B_{\max} = 32$	57.54±1.08	56.25±1.92	35.21±0.68	89.84±0.64	81.45±0.82
$B_{\max} = 128$	57.49±0.76	55.15±2.03	**38.93±0.24**	90.59±0.65	81.71±0.66
$s_{\min} = 5$	56.32±1.30	54.81±2.44	37.42±0.51	90.54±0.40	82.18±0.79
$s_{\min} = 10$	50.25±1.48	49.91±1.72	37.15±0.54	90.68±0.47	82.06±1.12
$n_{\min} = 1$	55.87±1.70	55.94±1.80	37.16±0.75	90.37±0.54	81.10±1.21
$n_{\min} = 16$	56.72±1.47	56.49±1.17	37.16±0.62	90.32±0.57	81.58±1.71
$n_{\min} = 64$	57.42±1.56	55.94±2.70	37.63±0.56	90.66±0.59	82.30±0.34
$\delta = 0.01$	57.49±0.76	55.24±2.04	37.39±0.51	90.71±0.59	82.22±0.86
$\delta = 0.10$	44.37±2.31	42.90±2.50	31.86±1.05	87.63±1.10	82.22±0.86
MLP+CE	44.24±2.62	43.50±3.02	31.98±0.98	88.42±0.62	73.90±1.80
Random grouping	43.31±1.40	42.08±2.88	32.30±0.85	87.46±1.07	71.45±3.29
Raw $\mathbf{X}$	44.37±2.31	42.90±2.50	31.86±1.05	87.63±1.10	73.37±1.13
$\mathbf{A}^2\mathbf{X}$	44.37±2.31	42.90±2.50	31.86±1.05	89.20±0.67	81.13±1.64

The best-performing setting varies across datasets. Increasing β from 0.5 to 1 raises mean accuracy on four datasets and lowers it on CoraFull.

With $B_{\max} = 128$, construction produces 52–53 balls on Cora, 63–66 on CiteSeer, 128 on CoraFull and Coauthor-CS, and 124–128 on Amazon-Photo. The larger budget leaves the ball-count range unchanged on Cora and expands it from 63–64 to 63–66 on CiteSeer. On CoraFull, accuracy improves over the default on all five seeds, with a mean paired gain of 1.53 ± 0.40 percentage points. This setting gives its highest mean accuracy and lowers the target Brier score from 0.6321 to 0.6026. On Coauthor-CS and Amazon-Photo, mean accuracy changes by -0.12 and -0.41 points, respectively.

Reducing the minimum child-ball size $n_{\min}$ from 32 to 1 changes mean accuracy by -1.61, $+0.70$, -0.23, -0.34, and -1.02 points on Cora, CiteSeer, CoraFull, Coauthor-CS, and Amazon-Photo, respectively. Allowing smaller child balls thus improves the mean only on CiteSeer.

Raising $s_{\min}$ requires more voting nodes before consensus can modify a target. At $s_{\min} = 10$, the proportions of training nodes with a positive consensus weight are 56.52%, 58.32%, 99.13%, 98.98%, and 90.03% in the same dataset order. On Cora and CiteSeer, more than 40% of training nodes retain their observed one-hot targets, and accuracy falls by more than 5 points (Table I.1).

Appendix I.2. Error reduction on training targets

Table I.2 reports the default configuration's five-seed mean decomposition of target Brier-score reduction under uniform 0.4 noise, as defined in Eq. (H.5).

Table I.2: Decomposition of the reduction in training-target Brier score under uniform 0.4 noise.

Dataset	One-hot label	Soft target	Reduction on corrupted labels	Error increase on correct labels	Net reduction
Cora	0.8118	0.4498	0.4034	0.0415	0.3620
CiteSeer	0.8162	0.5352	0.3304	0.0493	0.2811
CoraFull	0.8011	0.6321	0.1960	0.0269	0.1690
Coauthor-CS	0.8008	0.3130	0.5162	0.0285	0.4877
Amazon-Photo	0.8013	0.3664	0.4660	0.0310	0.4350

Error removed on corrupted labels exceeds error introduced on correct labels in all 25 dataset–seed runs. The five-seed mean reductions are 6.7–18.1 times the increases, with net reductions from 0.1690 on CoraFull to 0.4877 on Coauthor-CS. Random size-matched grouping also reduces Brier score but yields lower accuracy on all five datasets (Table I.1). Average target error alone therefore does not explain the accuracy advantage of representation-based grouping, consistent with the observation in Appendix H.2 that a target with a lower Brier score can still favor an incorrect class.

Nodes with a positive clean-vote margin ($\Delta_n > 0$) account for 75.64%, 65.02%, 36.13%, 88.93%, and 84.64% of training nodes on Cora, CiteSeer, CoraFull, Coauthor-CS, and Amazon-Photo, respectively. The net Brier-score reductions follow the same dataset order, while Table I.2 includes all training nodes.